Low-Degree Lower Bounds for Overcomplete Tensor Decomposition*

Abstract

We study a form of overcomplete tensor decomposition in the low-degree polynomial framework: given an order-k tensor with m components $v_1, \dots, v_m$ drawn from $\{\pm 1\}^n$, where $v_1^{\otimes k}$ is scaled up by a constant factor $\tau > 1$, the goal is for a polynomial of degree at most ℓ to recover the first entry of v_1, or the product of the first two entries if k is even. For $k \in \{3, 4\}$, degree $\ell^3 \le n$, and $m \ge Cn^{k/2}\ell^{k/2-1}$ for a large enough constant C, we prove that any polynomial of degree at most ℓ has negligible advantage, in mean square error, over the constant-zero estimator.

1 Definitions

Following Wein [1, Sections 1.1–1.2], we consider a form of tensor decomposition that fits the low-degree polynomial (LDP) framework: one component has larger norm, and the goal is to estimate its first entry. Let $[n] = \{1, \dots, n\}$, and write $v^A = \prod_{i \in A} v(i)$ for $A \subseteq [n]$. Fix $k \in \{3, 4\}$, integers $n \ge 4$ and $m \ge 2$, and $\tau > 1$. The observation is

$$T = \tau v_1^{\otimes k} + \sum_{j=2}^{m} v_j^{\otimes k}, \qquad v_1, \dots, v_m \overset{\text{iid}}{\sim} \mathrm{Unif}(\{\pm 1\}^n). \tag{1}$$

Here m denotes the rank parameter, namely the number of rank-1 terms; the first term has weight $\tau > 1$, and all remaining terms have weight one. Equivalently, the component vectors are $\tau^{1/k} v_1, v_2, \dots, v_m$, so estimating the first entry of the component of larger norm amounts to estimating $v_1(1)$, up to a known scale. Define

$$\Omega_k = \{I \subseteq [n] : 1 \le |I| \le k,\ |I| \equiv k \pmod 2\}, \qquad Y_I = \tau v_1^I + \sum_{j=2}^{m} v_j^I. \tag{2}$$

Each tensor entry equals Y_I for the set I of coordinates occurring an odd number of times, or, if $I = \varnothing$, the deterministic constant $m - 1 + \tau$. Conversely, every Y_I, $I \in \Omega_k$, occurs as a tensor entry. Thus polynomials in T and in $Y = (Y_I)_{I \in \Omega_k}$ generate the same observable polynomial space, with the same degree bound.

The target classes are

$$\mathcal{A}_3 = \{A \subseteq [n] : |A| = 1\}, \qquad \mathcal{A}_4 = \{A \subseteq [n] : |A| \in \{2, 4\}\}. \tag{3}$$

For $A \in \mathcal{A}_k$, put $x_A = v_1^A$ and $a = |A|$. For $k = 3$, taking $A = \{1\}$ gives the first-entry target $v_1(1)$. For $k = 4$, the tensor is unchanged by $v_1 \mapsto -v_1$, so we use sign-invariant targets; the pair target $v_1(1)v_1(2)$ was suggested by Wein [1, Section 2]. It is the first entry of the representative $v_1(2)v_1$, whose second entry is positive. The theorem also covers all pair and quartet targets in (3). For

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a positive integer ℓ, let $\mathbb{R}[T]_{\le \ell}$ be the space of real polynomials of degree at most ℓ in the tensor entries. Following Wein [1, Sections 1.2 and 1.4.1], define the *degree-ℓ minimum mean squared error* and the *degree-ℓ maximum correlation* by

$$\mathrm{MMSE}_{\le \ell}(x_A; T) = \inf_{f \in \mathbb{R}[T]_{\le \ell}} \mathbb{E}[(f(T) - x_A)^2], \tag{4}$$

$$\mathrm{Corr}_{\le \ell}(x_A; T) = \sup_{\substack{f \in \mathbb{R}[T]_{\le \ell} \\ \mathbb{E}[f(T)^2] > 0}} \frac{\mathbb{E}[x_A f(T)]}{\sqrt{\mathbb{E}[f(T)^2]}}. \tag{5}$$

Both optimizations range over all real polynomials in the full tensor entries. Since $\mathbb{E}[x_A^2] = 1$, the standard identity used in Wein [1, Fact 3.1] gives $\mathrm{MMSE}_{\le \ell}(x_A; T) = 1 - \mathrm{Corr}_{\le \ell}(x_A; T)^2$.

2 Statement

Theorem 1. *For every fixed $\tau > 1$, there exists $C_\tau < \infty$, independent of n, m, ℓ and uniform for $k \in \{3, 4\}$, such that*

$$\ell^3 \le n, \qquad m \ge C_\tau n^{k/2} \ell^{k/2-1} \tag{6}$$

imply, for every $A \in \mathcal{A}_k$,

$$\mathrm{Corr}_{\le \ell}(x_A; T)^2 \le n^{-|A|/2}, \qquad \mathrm{MMSE}_{\le \ell}(x_A; T) \ge 1 - n^{-|A|/2}. \tag{7}$$

3 Proof

We use Wein's reduction to the Fourier basis of the underlying Rademacher variables [1, Sections 1.4.1 and 3.1–3.2], expressed below through a moment-matching witness. Throughout the proof, C denotes a universal constant that may change between occurrences. Constants with subscript τ depend only on τ. Fix an allowed target A, and abbreviate $\Omega = \Omega_k$. For uniform $z \in \{\pm 1\}^n$, set

$$F(t) = \mathbb{E}_z \exp\left(\sum_{I \in \Omega} t_I z^I\right), \qquad \phi_B(t) = \frac{\mathbb{E}_z[z^B \exp(\sum_{I \in \Omega} t_I z^I)]}{F(t)} \quad (\varnothing \ne B \subseteq [n]). \tag{8}$$

These functions initially define holomorphic germs at zero, with $\phi_B(0) = 0$. Let $K_I(t)$ be the joint cumulant of $(z_i)_{i \in I}$ under this tilt, for $I \in \Omega$. For $k = 3$, its coordinates are

$$K_i = \phi_i, \qquad K_{ijs} = \phi_{ijs} - \phi_i\phi_{js} - \phi_j\phi_{is} - \phi_s\phi_{ij} + 2\phi_i\phi_j\phi_s. \tag{9}$$

For $k = 4$, global spin-flip symmetry makes every odd tilted moment vanish, so

$$K_{ij} = \phi_{ij}, \qquad K_{ijsr} = \phi_{ijsr} - \phi_{ij}\phi_{sr} - \phi_{is}\phi_{jr} - \phi_{ir}\phi_{js}. \tag{10}$$

All indices within a displayed set are distinct. Character orthogonality gives $K(0) = 0$ and $DK(0) = \mathrm{Id}$. For $t \in \mathbb{C}^\Omega$, write

$$\|t\|_{\mathrm{loc}} = \max_{i \in [n]} \sum_{\substack{I \in \Omega \\ i \in I}} |t_I|. \tag{11}$$

Lemma 1. *There exists $\rho_\tau > 0$, independent of n, k, A, such that*

$$\mathcal{C}_A(q) = \phi_A(\tau K^{-1}(q)) \tag{12}$$

is holomorphic on a neighborhood of the closed ball $\|q\|_{\mathrm{loc}} \le \rho_\tau$ and satisfies $|\mathcal{C}_A(q)| \le 1$ there.

Proof. Write $K(t) = \sum_{r\geq 1} T_r(t,\dots,t)$. Differentiating tilted cumulants gives the symmetric multilinear coefficients

$$T_{r,I}(w_1,\dots,w_r) = \frac{1}{r!}\sum_{J_1,\dots,J_r\in\Omega} \kappa_0\big((z_i)_{i\in I}, z^{J_1},\dots,z^{J_r}\big)\prod_{b=1}^{r}(w_b)_{J_b}. \tag{13}$$

We use the dependency-graph inequality [2, Eq. (47)]

$$|\kappa(X_1,\dots,X_s)| \leq 2^{s-1}\,\mathrm{ST}(G) \tag{14}$$

for variables bounded in absolute value by one with dependency graph G, where $\mathrm{ST}(G)$ is the number of spanning trees. For (13), join character occurrences whose supports intersect. This is a dependency graph, including when occurrences repeat.

Fix an output coordinate i and output size $p \in \{k-2, k\}$. Root a spanning tree at the marked spin z_i. An interaction child b attached to a parent support B contributes at most

$$\sum_{J: J\cap B\neq\varnothing} |(w_b)_J| \leq |B|\|w_b\|_{\mathrm{loc}} \leq k\|w_b\|_{\mathrm{loc}}.$$

Each of the other $p-1$ marked spin children has at most k coordinate choices. These subtree bounds are uniform in the parent's support. Summing ordered choices of marked coordinates overcounts the output sets. Thus Cayley's formula and (14) give

$$\sum_{\substack{I\ni i\\ |I|=p}} |T_{r,I}(w_1,\dots,w_r)| \leq \frac{(2k)^{r+p-1}(r+p)^{r+p-2}}{r!}\prod_{b=1}^{r}\|w_b\|_{\mathrm{loc}}. \tag{15}$$

Since $k, p \leq 4$, the sum of the two numerical prefactors is at most C_0^r for a universal C_0. Consequently

$$\|T_r(w_1,\dots,w_r)\|_{\mathrm{loc}} \leq C_0^r\prod_{b=1}^{r}\|w_b\|_{\mathrm{loc}}. \tag{16}$$

The resulting series converges on a dimension-independent complex ball. To identify it with the actual moment ratios, let L_r be the homogeneous degree-r part of the formal logarithm of F. Here $L_1 = 0$. Rooting the interaction-only tree at its first interaction and using $\sum_I |t_I| \leq n\|t\|_{\mathrm{loc}}$ gives, for $r \geq 2$,

$$|L_r(t)| \leq \frac{2^{r-1}r^{r-2}}{r!}n\|t\|_{\mathrm{loc}}(k\|t\|_{\mathrm{loc}})^{r-1} \leq nC^r\|t\|_{\mathrm{loc}}^r. \tag{17}$$

The logarithmic series converges on a universal ball. The germ identity $F = \exp(\sum_r L_r)$ extends throughout that connected ball, so F has no zeros there. The series for K therefore agrees with (9)–(10).

Set $H = K - \mathrm{Id}$. Since $T_1 = \mathrm{Id}$, the mixed bound (16) implies

$$\|DH(t)\|_{\mathrm{loc}\to\mathrm{loc}} \leq \sum_{r\geq 2} rC_0^r\|t\|_{\mathrm{loc}}^{r-1}. \tag{18}$$

Choose a universal $\eta > 0$ so that the right-hand side is at most $1/2$ on the closed η-ball, inside the analytic domain. For $\|q\|_{\mathrm{loc}} < \eta/2$, the map $t \mapsto q - H(t)$ contracts that ball. Its holomorphic iterates converge uniformly to the inverse, and

$$\|K^{-1}(q)\|_{\mathrm{loc}} \leq 2\|q\|_{\mathrm{loc}}, \qquad \|K(t)\|_{\mathrm{loc}} \leq \tfrac{3}{2}\|t\|_{\mathrm{loc}} \quad (\|t\|_{\mathrm{loc}} \leq \eta). \tag{19}$$

For singleton targets when $k = 3$ and pair targets when $k = 4$, ϕ_A is a coordinate of K. For a quartet,

$$\phi_{ijsr} = K_{ijsr} + K_{ij}K_{sr} + K_{is}K_{jr} + K_{ir}K_{js}.$$

Shrinking η, and then taking a sufficiently small $\rho_\tau < \eta/(4\tau)$, proves the asserted holomorphy and bound for $\mathcal{C}_A$. □

For a multiset S of elements of Ω, write

$$|S| = \sum_I S_I, \qquad S! = \prod_I S_I!, \qquad q^S = \prod_I q_I^{S_I},$$

and let $\triangle S$ be the symmetric difference of its elements, with multiplicity. Expand

$$\mathcal{C}_A(q) = \sum_{d\geq 1} \mathcal{C}_{A,d}(q), \qquad \mathcal{C}_{A,d}(q) = \sum_{|S|=d} \frac{c_{A,S}}{S!} q^S, \qquad \mathcal{E}_{A,d} = \sum_{|S|=d} \frac{c_{A,S}^2}{S!}. \tag{20}$$

The coefficients are real.

Lemma 2. *For all $d \geq 1$,*

$$\mathcal{E}_{A,d} \leq n^{-a/2} \big(B_\tau n^{k/2} d^{k/2-1}\big)^d \exp(4d^4/n), \tag{21}$$

where B_τ is independent of n, k, A, d.

Proof. Coordinate-sign changes commute with K and its unique local inverse, and multiply the response by the sign character indexed by A. Hence

$$c_{A,S} = 0 \quad \text{unless} \quad \triangle S = A. \tag{22}$$

Equivalently, S together with A forms an *even cover*: every coordinate occurs an even number of times, counting multiplicities. This is the even-cover restriction used by Wein [1, Lemma 3.16], with the singleton target there replaced by A. In particular, contributing degrees satisfy $kd \equiv a \pmod 2$. For $k = 3$ this requires odd d; for $k = 4$ it imposes no degree restriction.

For a fixed multiset S, set $s_i = \sum_{I\ni i} S_I$ and $M_I = \max_{i\in I} s_i$. On the active variables, choose radii $r_I = \rho_\tau S_I/M_I$. Since $\sum_{I\ni i} r_I \leq (\rho_\tau/s_i)\sum_{I\ni i} S_I = \rho_\tau$ for every occurring coordinate i, this polydisc lies in the ball of Lemma 1. Cauchy's estimate yields

$$|c_{A,S}| \leq S! \prod_I r_I^{-S_I} \leq \rho_\tau^{-d} \prod_I M_I^{S_I} \leq \rho_\tau^{-d} \prod_{i:s_i>0} s_i^{s_i}. \tag{23}$$

Here we used $S_I! \leq S_I^{S_I}$ and $M_I \leq \prod_{i\in I} s_i$.

We refine Wein's counting of even covers [1, Lemma 3.17] by keeping track of coordinate multiplicities. Passing from multisets to ordered tuples gives

$$\mathcal{E}_{A,d} \leq \frac{\rho_\tau^{-2d}}{d!} \sum_{\substack{I_1,\dots,I_d\in\Omega \\ \triangle_b I_b = A}} \prod_{i:s_i>0} s_i^{2s_i}. \tag{24}$$

Fix one of the at most 2^d choices of input sizes in $\{k-2, k\}$. Let $s \leq kd$ be its number of incidence slots. A contributing pattern has $s \geq a$ and $s \equiv a \pmod 2$; put $L = (s-a)/2$. The multiplicities can be written as

$$s_i = 1 + 2b_i \quad (i \in A), \qquad s_j = 2 + 2b_j \quad (j \notin A \text{ occurring}).$$

If h nonroot coordinates occur and $e = \sum_i b_i$, then $h = L - e$. There are at most $n^h/h!$ supports and $\binom{e+h+a-1}{e} = \binom{L+a-1}{e}$ weak compositions of e.

For fixed multiplicities, there are $s!/\prod_i s_i!$ assignments to the labeled slots. Allowing repetitions within an input set only enlarges the count. If $u = b + 2b' \le 4d$, where $b \in \{1, 2\}$, then

$$\frac{u^{2u}}{u!} \le \exp(u)u^u \le (16\exp(1))^u d^{2b'}. \tag{25}$$

Indeed $u^u \le (4d)^{2b'}u^2$, $4^{2b'} \le 4^u$, and $u^2 \le 4^u$. Thus the fixed size pattern contributes at most

$$\frac{C_\tau^d s!}{d!} \sum_{e=0}^{L} \frac{n^{L-e}}{(L-e)!} \binom{L+a-1}{e} d^{2e}. \tag{26}$$

If the support size exceeds the available coordinates, its actual contribution is zero, so the relaxed support bound remains valid.

After factoring out $n^L/L!$, use

$$\frac{L!}{(L-e)!}\binom{L+a-1}{e} d^{2e} \le \frac{[L(L+a-1)d^2]^e}{e!} \le \frac{(4d^4)^e}{e!}. \tag{27}$$

The last inequality follows from $L \le (3d-1)/2$, $2d-1$, or $2d-2$ for $(k, a) = (3, 1), (4, 2), (4, 4)$, respectively. When $L = 0$, only $e = 0$ occurs and the expression is one. Summing in (26) therefore gives

$$\frac{C_\tau^d s! n^L}{d!L!} \exp(4d^4/n). \tag{28}$$

Increasing s by two multiplies $s!n^{(s-a)/2}/((s-a)/2)!$ by $n(s+2)(s+1)/(L+1) \ge 1$. Since $kd \equiv a \pmod 2$, the maximum occurs at $s = kd$. Put $L_* = (kd-a)/2$. Except for $(k, a, d) = (4, 4, 1)$, which is immediate, $L_* \ge d$, and elementary factorial bounds give

$$\frac{(kd)!}{d!L_*!} \le C^d d^{kd-d-L_*} = C^d d^{(k/2-1)d+a/2} \le C^d d^{(k/2-1)d}.$$

In the last inequality the constant is enlarged using $d^{a/2} \le d^2 \le 4^d$. Now $n^{L_*} = n^{-a/2}n^{kd/2}$; summing the at most 2^d size patterns proves (21). □

Define the auxiliary index sets

$$\Lambda_3 = \{B \subseteq [n] : 1 \le |B| \le 3\}, \qquad \Lambda_4 = \Omega_4.$$

Following Wein's use of Fourier monomials with distinct component indices [1, Section 3.4], we average over distinct background components. For a multiset $R = (B_1, \dots, B_r)$ from Λ_k, with $r \le m-1$, let

$$U_R = \frac{1}{(m-1)^{\underline{r}}} \sum_{\substack{j_1,\dots,j_r \in \{2,\dots,m\} \\ \text{all distinct}}} \prod_{b=1}^{r} v_{j_b}^{B_b}, \qquad (m-1)^{\underline{r}} = (m-1)(m-2)\cdots(m-r). \tag{29}$$

Set $U_\varnothing = 1$. For a polynomial $P(u) = \sum_R p_R u^R$ of degree at most $m-1$, define $\mathrm{Lift}(P) = \sum_R p_R U_R$. The auxiliary pair characters used for $k = 3$ are functions of the underlying Rademacher variables, not additional observations.

Let $b_k(u)$ be obtained from (9) or (10) by replacing each ϕ_B with the independent letter u_B. Then

$$b_k(\phi(t)) = K(t), \qquad \deg b_k \le h_k, \qquad h_3 = 3, \quad h_4 = 2. \tag{30}$$

When $m - 1 \ge h_k \ell$, set

$$P_{A,\ell}(u) = \sum_{d=1}^{\ell} \mathcal{C}_{A,d}(b_k(u)), \qquad W_{A,\ell} = \mathrm{Lift}(P_{A,\ell}). \tag{31}$$

Lemma 3. *Suppose $m - 1 \ge 6\ell$. Then $W_{A,\ell}$ satisfies*

$$\mathbb{E}[W_{A,\ell} f(Y)] = \mathbb{E}[x_A f(Y)] \qquad (\deg f \le \ell), \tag{32}$$

and, for $1 \le d \le \ell$,

$$\left\| \mathrm{Lift}[\mathcal{C}_{A,d}(b_k(u))] \right\|_2 \le 6^d \sqrt{\frac{\mathcal{E}_{A,d}}{(m-1)^{\underline{d}}}}. \tag{33}$$

Proof. Products of the underlying independent Rademacher variables form the standard Fourier basis for functions on the hypercube. Every distinct basis monomial in U_R occurs $R!$ times, where $R! = \prod_B R_B!$, and there are $(m-1)^r/R!$ such monomials. Different multisets have disjoint Fourier supports. Consequently

$$\langle U_R, U_{R'} \rangle = 0 \quad (R \ne R'), \qquad \|U_R\|_2^2 = \frac{R!}{(m-1)^{\underline{|R|}}}. \tag{34}$$

Independence gives

$$\mathbb{E}[U_R e^{\langle t, Y \rangle}] = F(\tau t) F(t)^{m-1} \phi(t)^R, \tag{35}$$

$$\mathbb{E}[x_A e^{\langle t, Y \rangle}] = F(\tau t) F(t)^{m-1} \phi_A(\tau t). \tag{36}$$

Since $K(0) = 0$ and $\mathcal{C}_A(K(t)) = \phi_A(\tau t)$, the polynomial in (31) satisfies

$$P_{A,\ell}(\phi(t)) = \phi_A(\tau t) + O_{\deg}(\ell + 1).$$

Its degree in the auxiliary variables is at most $h_k \ell \le 3\ell \le m - 1$. Multiplying the formal identity by $F(\tau t) F(t)^{m-1}$ and comparing Taylor coefficients in (35)–(36) proves (32).

For the norm bound, let $V = \mathbb{R}^\Omega$ and $H = \mathbb{R}^{\Lambda_k}$, with standard orthonormal bases. Represent the homogeneous parts of b_k by symmetric tensor maps $Q_j : V \to H^{\otimes j}$:

$$(b_k)_I^{[j]}(u) = \langle Q_j e_I, u^{\otimes j} \rangle.$$

The linear map is the basis inclusion. For $k = 3$, a triple output has six ordered singleton–pair coefficients $-1/2$ in Q_2, and six ordered three-singleton coefficients $1/3$ in Q_3. For $k = 4$, a quartet output has six ordered complementary-pair coefficients $-1/2$ in Q_2. Within each map, distinct output sets have disjoint tensor supports, because the union of an input tuple identifies its output. Thus

	$\|Q_1\|$	$\|Q_2\|$	$\|Q_3\|$
$k = 3$	1	$\sqrt{3/2}$	$\sqrt{2/3}$
$k = 4$	1	$\sqrt{3/2}$	0

and $\displaystyle\sum_{j=1}^{h_k} \|Q_j\| \le 6.$ (37)

Write $\mathcal{C}_{A,d}(q) = \langle M_d, q^{\otimes d} \rangle$, with M_d symmetric. Its entry on an ordered tuple of multiset S is $c_{A,S}/d!$, so

$$d! \|M_d\|_{\mathrm{HS}}^2 = \mathcal{E}_{A,d}. \tag{38}$$

For any symmetric order-s tensor M, (34) implies the exact identity

$$\left\|\mathrm{Lift}\langle M, u^{\otimes s}\rangle\right\|_2^2 = \frac{s!}{(m-1)^{\underline{s}}}\|M\|_{\mathrm{HS}}^2. \tag{39}$$

For one substitution choice $(j_1, \ldots, j_d)$, the resulting tensor has order $s = \sum_b j_b \in [d, h_k d]$ and norm at most $\prod_b \|Q_{j_b}\| \cdot \|M_d\|_{\mathrm{HS}}$. Orthogonal symmetrization leaves its polynomial unchanged and cannot increase the norm. When $m - 1 \geq 6\ell$, the successive ratio

$$\frac{(s+1)!/(m-1)^{\underline{s+1}}}{s!/(m-1)^{\underline{s}}} = \frac{s+1}{m-1-s} \leq 1 \qquad (d \leq s < 3d)$$

shows that $s!/(m-1)^{\underline{s}} \leq d!/(m-1)^{\underline{d}}$ throughout the needed range. Using (38)–(39) and summing all substitution choices by the triangle inequality gives

$$\left\|\mathrm{Lift}[\mathcal{C}_{A,d}(b_k(u))]\right\|_2 \leq \left(\sum_{j=1}^{h_k} \|Q_j\|\right)^d \sqrt{\frac{\mathcal{E}_{A,d}}{(m-1)^{\underline{d}}}},$$

which proves (33). □

Proof of Theorem 1. For $d \leq \ell$, the assumption $\ell^3 \leq n$ gives $d^4/n \leq d$. Absorbing the exponential in Lemma 2 into its constant, we obtain

$$\mathcal{E}_{A,d} \leq n^{-a/2}\left(B_\tau n^{k/2} d^{k/2-1}\right)^d. \tag{40}$$

Choose $C_\tau \geq \max\{8, 288B_\tau\}$. Under (6), $n^{k/2}\ell^{k/2-1} \geq \ell$, so $m - 1 \geq 6\ell$ and $m - \ell \geq m/2$. By Lemma 3, the triangle inequality across degrees, and $(m-1)^{\underline{d}} \geq (m-\ell)^d$,

$$\|W_{A,\ell}\|_2 \leq n^{-a/4} \sum_{d=1}^{\ell} \left(\frac{36B_\tau n^{k/2} d^{k/2-1}}{m-\ell}\right)^{d/2}. \tag{41}$$

Each base is at most

$$\frac{72B_\tau n^{k/2}\ell^{k/2-1}}{m} \leq \frac{1}{4},$$

and hence $\|W_{A,\ell}\|_2 \leq n^{-a/4}\sum_{d\geq 1} 2^{-d} = n^{-a/4}$. For every polynomial f of degree at most ℓ, moment matching yields

$$\mathbb{E}[(f(Y) - x_A)^2] = 1 - \|W_{A,\ell}\|_2^2 + \mathbb{E}[(f(Y) - W_{A,\ell})^2] \geq 1 - n^{-a/2}.$$

Alternatively, Cauchy–Schwarz applied to $\mathbb{E}[x_A f(Y)] = \mathbb{E}[W_{A,\ell} f(Y)]$ gives the squared-correlation bound directly. The equivalence of the T- and Y-polynomial spaces completes the proof. □

References

[1] Alexander S. Wein. Average-case complexity of tensor decomposition for low-degree polynomials. In *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC 2023)*, pp. 1685–1698, 2023. doi:10.1145/3564246.3585232. arXiv:2211.05274v2.

[2] Valentin Féray, Pierre-Loïc Méliot, and Ashkan Nikeghbali. *Mod-ϕ convergence, I: Normality zones and precise deviations.* arXiv:1304.2934v4, 2015.